IntelliNova

Your class material, turned into a tutor that shows its work.

Cited answers link to the exact page, slide or video moment they came from, and anything else is clearly labelled. Quizzes adapt to each student, and it plans their revision before they forget. It runs entirely on the school's own computer.

Grounded

Answers come from the uploaded textbooks, slides and lectures, with clickable citations. Off-syllabus questions are declined or clearly flagged.

Adaptive

Per-topic mastery (Bayesian knowledge tracing), forgetting curves (FSRS) and difficulty targeting (IRT/Elo) drive every quiz and plan.

Private & free to run

Open models run locally through Ollama. No cloud AI, no per-question fees, and student data stays on the machine.

01 · THE PROBLEM

Students need help that matches their class, not the whole internet

A Class 10 student revising Electricity at 10 pm has two options: re-read a 40-page chapter, or ask a general AI chatbot. Neither is built around what their teacher actually taught or what their exam will test.

TRUST

General chatbots make things up

They answer confidently from the open internet. Answers can contradict the textbook, use a different syllabus, or be wrong, and there is no way to check where a claim came from.

FIT

One-size practice wastes time

Question banks don't know which topics a student has mastered and which they're forgetting. Strong students repeat easy questions; struggling students hit walls.

MEMORY

Learning fades before the exam

Without spaced revision, most of a topic is forgotten within weeks. Students don't know *when* to revise *what*, especially with a fixed exam date.

LANGUAGE

English material, Hindi thinking

Many Indian students learn from English textbooks but understand best in Hindi or Hinglish. Most tools work in one language only.

FORMAT

Material is scattered across formats

Knowledge sits in PDFs, slide decks, recorded lectures and board diagrams. Turning it into something searchable normally takes manual work nobody has time for.

COST & PRIVACY

Cloud AI is costly and sends data away

Per-question API fees add up across a school, and student questions and performance data leave the institution.

Our answer

IntelliNova turns a school's own material into a tutor, quiz engine and revision planner. Teachers upload textbooks, slides and lecture videos. The system organises them into topics and links every piece of content to its source location. Students then learn, ask, practise and revise against that material only. A learner model tracks what each student knows and how fast they forget, and an evaluation harness measures how accurate the whole system is.

> **Who it is for:** schools and coaching centres (the admin console) and their students (the student app). Built for Indian curricula: CBSE, ICSE, State boards and IB/IGCSE, Classes 9–12 and early college, in English, Hindi and Hinglish.

02 · WHAT WE BUILT

Two connected apps on one local AI platform

A **student app** for learning and an invite-only **admin console** for the school. Both run on one platform that ingests material, answers questions, generates and checks quizzes, models each learner and evaluates itself. The two are strictly separate: student accounts can't open the console, and staff accounts can't use the student app.

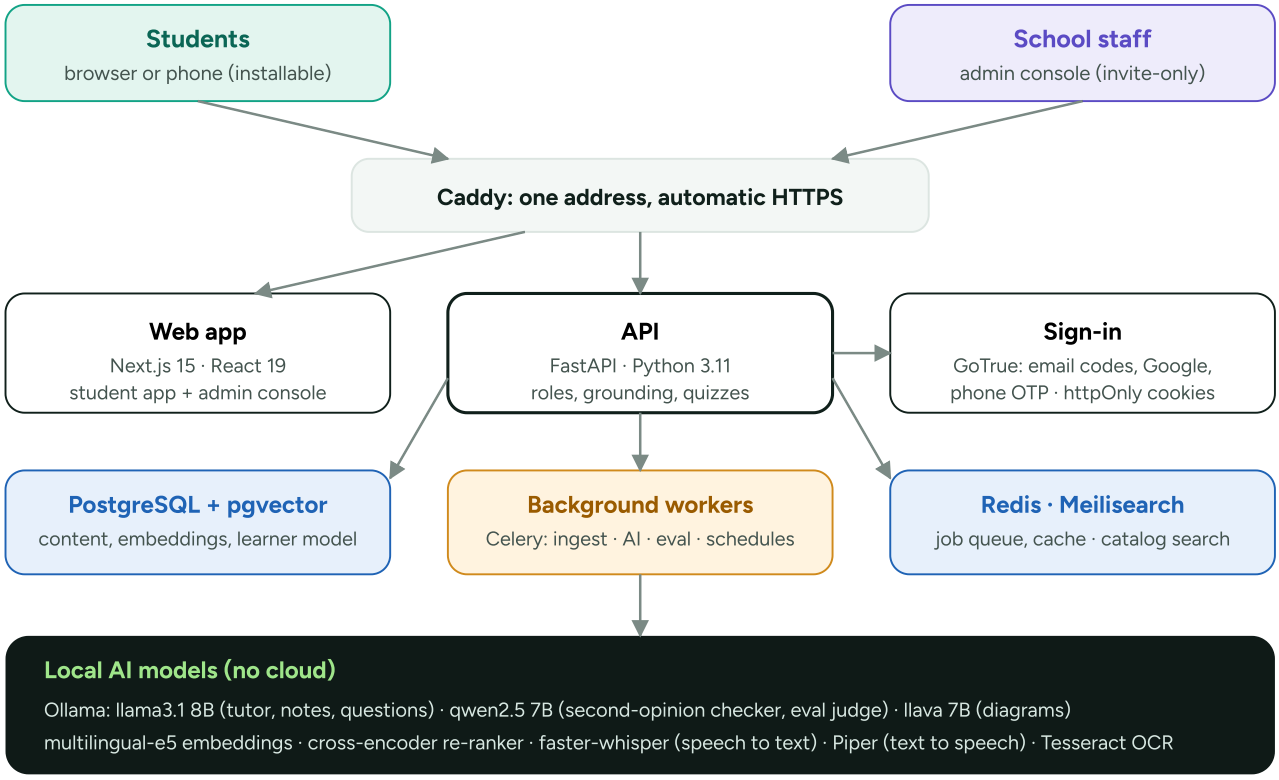
STUDENT APP	WHAT THE STUDENT GETS
Onboarding	Board, class, subjects and language, then an optional diagnostic quiz that sets their starting level on each topic.
Home	Today's plan, topics that need attention, where to continue, streaks and reminders.
Learn	Chapters and topics with the class material, curated and auto-found videos, and a *flow map* of prerequisites coloured by mastery.
Ask Tutor	Answers from the class material with each sentence cited; anything else is clearly labelled. Tapping a citation opens the exact page, slide or timestamp. English, Hinglish or Hindi; typed or spoken.
Practice	MCQ, numerical and short-answer quizzes and mock exams on a chosen scope. Difficulty adapts per question, with cited feedback and a weak-topic and misconception report.
Study AI	Notes, flashcards, mind maps, audio briefs and `.pptx` revision slides, generated from the student's own uploads or the class material.
Study Plan	Exam date plus minutes per day gives a day-by-day schedule: new topics in course order, weak topics first, reviews timed to the forgetting curve, and a final revision window.
Also	Exam Prep (entrance-exam syllabus and facts), Career Explorer, My Progress, Saved and Search.

ADMIN CONSOLE	WHAT THE SCHOOL CONTROLS
Curriculum	Subjects, chapters, topics and prerequisites, with versioned publishing. Students see only published content.
Knowledge base	Upload PDFs, slide decks and videos or YouTube links, and watch them process. Review low-confidence tags and accept suggested new topics.
Resources · Exams · Careers	Curated links and videos, exam structures with verified facts and official sources, and career data and eligibility rules.
Generated content	Review AI-written questions and notes before students rely on them.
Evaluation	Test sets, question bank, RAGAS / DeepEval / TruLens runs, per-question results and downloadable reports.
Users · Analytics	Invite curators and admins, suspend accounts, handle privacy requests, and see class-wide weak topics and searches.

03 · HOW IT WORKS

Architecture: everything runs on one machine

One `docker compose up` starts the whole system. The AI models run locally through Ollama on the school's GPU, or on the CPU if there is no GPU.



System architecture. Every box is a container on the same machine. The API handles quick requests; slow work (processing a lecture, running an evaluation) goes to background workers.

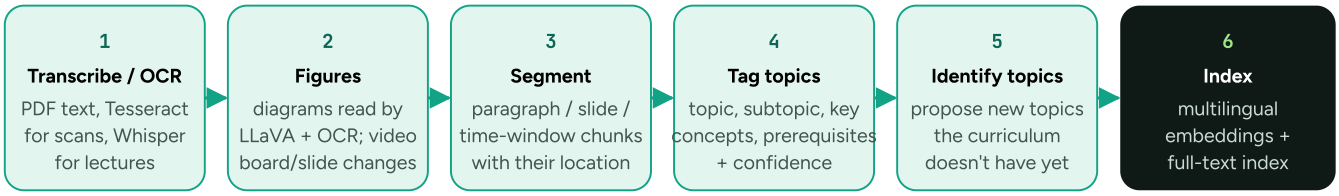
Design choices that matter

- **Source-linked chunks are the unit of everything.** Every piece of content keeps its origin (page, slide number, video timestamp, figure). Answers, quiz questions, notes and feedback all cite those chunks, so any claim can be traced back.
- **Two models check each other.** A second, different model (qwen2.5) verifies generated questions and judges answers, so one model's blind spots are less likely to slip through.
- **Hybrid retrieval in the database.** Vector similarity (pgvector) and keyword search (Postgres full-text) are merged with reciprocal-rank fusion, then re-ranked by a multilingual cross-encoder. Exact terms like "Ohm's law" and paraphrases like "relation of current and voltage" both work.
- **Local by default.** Sized for a 16 GB laptop with an 8 GB GPU. Bigger models can be swapped in with one setting.

04 · HOW IT WORKS: KNOWLEDGE BASE

From raw files to a source-linked knowledge base

A teacher uploads a textbook PDF, a slide deck, a lecture video or a YouTube link. No manual preprocessing: six automatic stages turn it into tagged, searchable units.



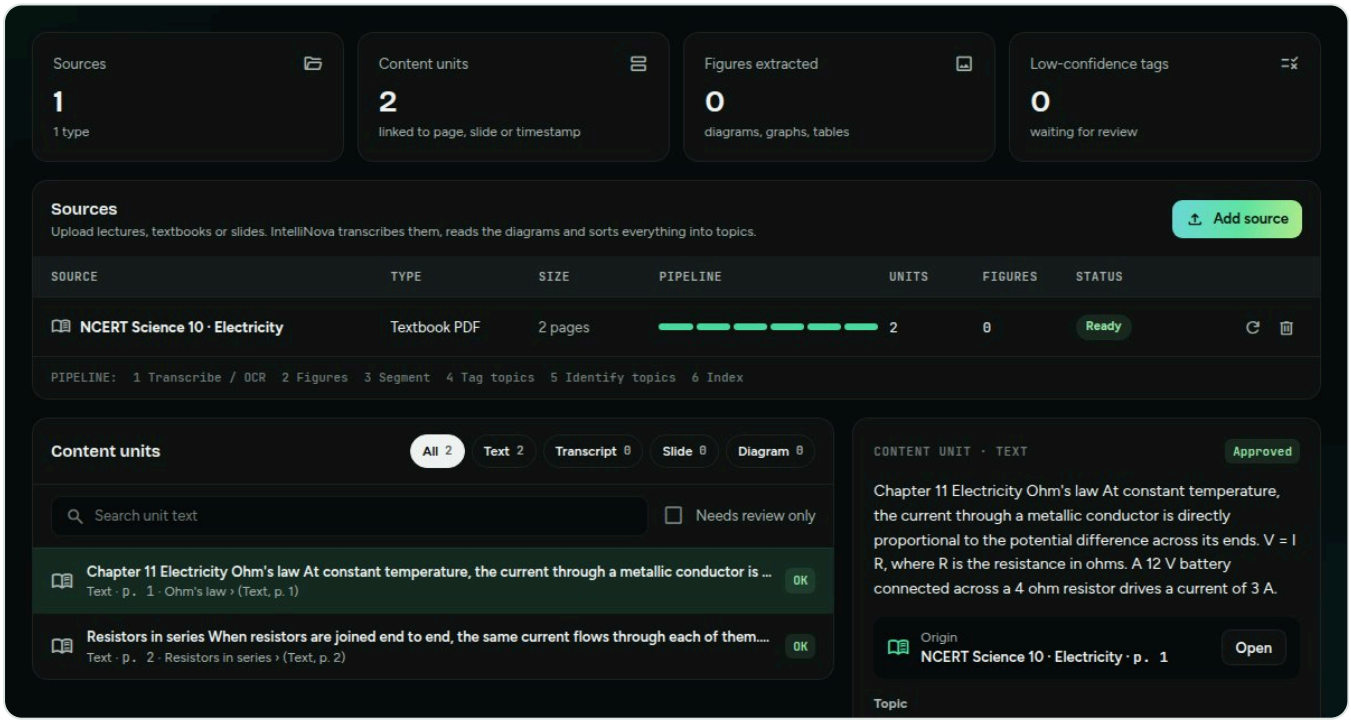
Every resulting unit keeps its origin: "p. 2", "Slide 4", "Video 1 · 12:34" or "p. 5 · Fig. 11.2"

Any format

PDFs (PyMuPDF, optionally layout-aware Docling; scanned pages OCR'd in English and Hindi), PowerPoint (one unit per slide plus embedded images), uploaded videos (Whisper transcript, plus OCR of board and slide changes found with OpenCV) and YouTube (captions when available).

Humans stay in control

Low-confidence tags wait for staff review. Material the curriculum doesn't cover appears under *Suggested topics*; one click adds it as a draft chapter or topic with its subtopics. Nothing reaches students until it's published.



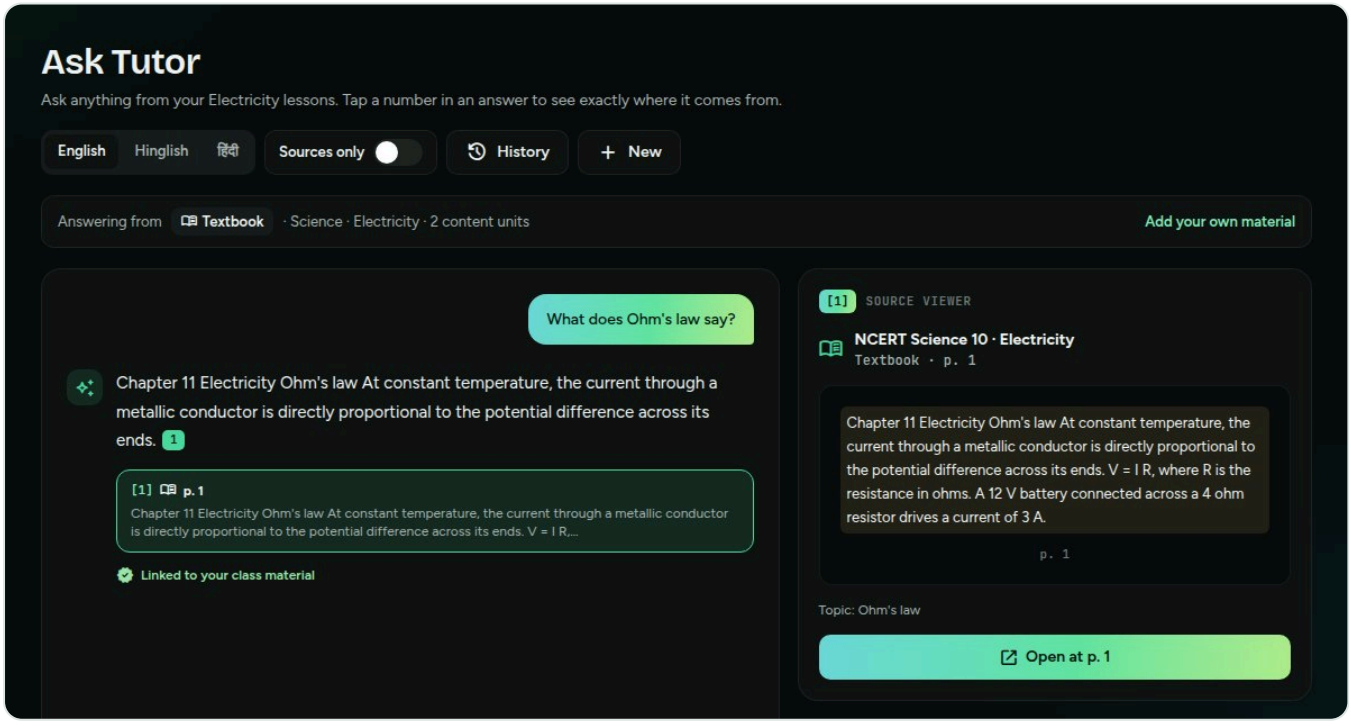
Admin › Knowledge base: the six-stage pipeline per source, and each content unit with its origin (page), topic, subtopic, key concepts and prerequisites.

05 · HOW IT WORKS: THE TUTOR

Answers that show their work, or honestly say "not in your material"

The tutor is built so it *cannot* quietly fall back on internet knowledge. Every answer is either backed by cited class material, declined, or clearly marked as outside the material.

1. **Understand the question.**
   Follow-ups such as "and in parallel?" are rewritten into a standalone question using the conversation so far.
2. **Retrieve.**
   Hybrid search (vectors + keywords, fused) finds the best passages from the published class material and the student's own uploads. A cross-encoder re-orders them by relevance.
3. **Answer only from numbered excerpts.**
   The model gets the passages as [1], [2], … and must cite an excerpt number after every sentence. If they don't contain the answer, it must say so.
4. **Check faithfulness.**
   Each sentence is checked against the excerpt it cites. If fewer than 60% are supported, the answer is thrown away rather than shown.
5. **Show it with citations.**
   Citations open a source viewer at the exact page, slide or timestamp, with the passage highlighted and an *Open at p. 2* button.
6. **Otherwise decline or flag.**
   In *Sources only* mode it declines ("Your class material doesn't cover this, so I won't guess"). Otherwise it gives a short general explanation in a separate box clearly labelled as outside the class material.



Ask Tutor: the answer carries citation [1]; the source viewer shows the exact passage and opens the textbook at p. 1. Language switch (English / Hinglish / Hindi), Sources-only switch, voice input and hands-free voice mode are in the toolbar.

Quizzes that are correct, never repeat, and adapt to each student

Generating questions that can be trusted

1. **Generate from the material.**
   For a topic, the model writes MCQ, numerical and short-answer questions from that topic's source-linked passages. Each question records its topic, source location and difficulty, plus the misconception behind each wrong option.
2. **Verify twice.**
   Numerical answers are re-computed with a symbolic maths solver (SymPy) from the stated expression. A *second, different model* answers every question independently. If the two disagree, the question is held back.
3. **Remove near-duplicates.**
   Each new question is compared with the topic's question bank by embedding similarity; paraphrases of existing questions are dropped.
4. **Never repeat for a student.**
   Questions a student has already seen are skipped in later quizzes and mock exams.

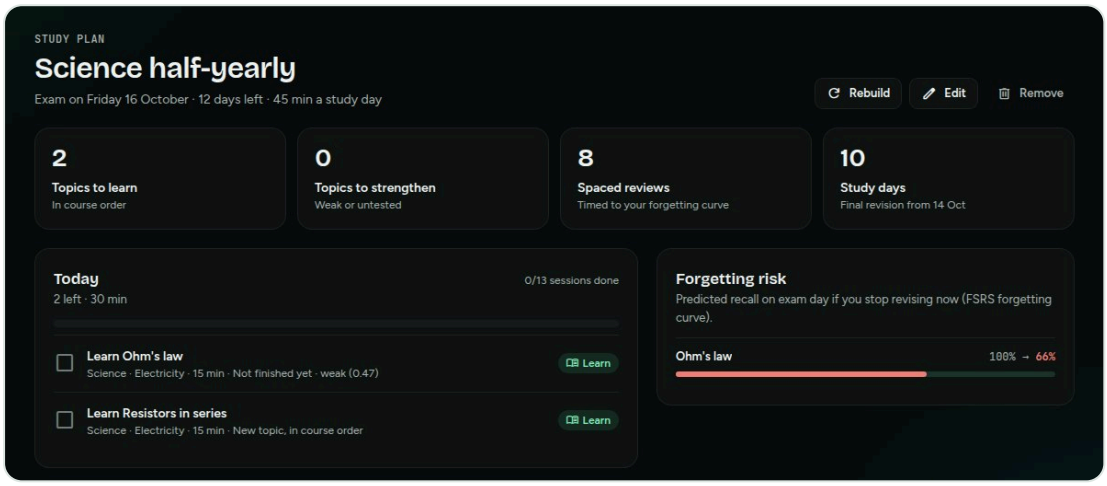
Modelling what each student knows

TECHNIQUE	WHAT IT DOES IN INTELLINOVA
Bayesian Knowledge Tracing	Keeps a probability that the student has mastered each topic. It updates after every quiz answer and, more softly, from tutor conversations. Status is Weak below 0.5, Developing up to 0.8, then Mastered. Parameters can be refitted from real data with pyBKT.
IRT / Elo difficulty	Estimates the student's ability and each question's difficulty, and picks the next question at about a 70% chance of success: challenging but not discouraging. Question difficulty is updated from real answers.
FSRS spaced repetition	Predicts how likely the student is to recall a topic today and on exam day. Reviews are scheduled just before recall drops to 90%.
Diagnostic & intake	New students take a short diagnostic quiz (or skip it), and the tutor's first chat asks which chapters they've covered and which topic feels hardest. Together they give the model a starting point instead of a blank slate.

After every quiz the student sees cited feedback on each answer, how their mastery changed per topic, their weak topics, and likely misconceptions inferred from the wrong options they chose. One click starts targeted practice, a tutor conversation or flashcards on exactly those topics.

From "what should I study?" to a day-by-day plan

The learner model feeds directly into planning. A student enters an exam date and the minutes they can study each day. IntelliNova lays out every day up to the exam and rebuilds the plan each morning, so missed work rolls forward.



Study Plan: topics to learn (in course order), weak topics to strengthen, spaced reviews timed to the forgetting curve, a final-revision window and a forgetting-risk panel predicting recall on exam day.

6a

Course flow map

Every topic of a subject in teaching order with "must know first" arrows, each coloured by the student's mastery, so a student can see what to learn next and why.

6b

Revision packs for weak topics

One click from the dashboard builds notes, flashcards, a mind map, a revision checklist, an audio brief and downloadable `.pptx` slides, all from the approved class material for the student's weakest topics, and all cited.

6c

Plans that respect forgetting

New topics are followed by practice the next day and reviews about 3 and 7 days later. Mastered topics are reviewed when FSRS says recall is slipping. The last days rank topics by predicted recall on exam day, ending with a mixed mock test.

6d

English material, Hindi explanations

Multilingual embeddings mean a question in Hindi or Hinglish finds the right passage in an English textbook. The tutor can answer in English, Hinglish or Hindi, and OCR reads Hindi textbooks.

6e

Audio tutoring

Spoken questions are transcribed on the server by Whisper, so it works in every browser and nothing leaves the machine. Answers and audio briefs are read aloud by Piper voices (English and Hindi). Hands-free mode detects when the student stops talking.

08 · HOW WE KNOW IT WORKS

A built-in evaluation harness

The school can measure accuracy itself, on its own material, whenever the content or models change. Admin › Evaluation runs the full pipeline against a held-out test set and scores it with a standard framework.

METRIC	WHAT IT ASKS	TARGET
Faithfulness	Is every claim in the answer supported by the retrieved passages?	0.85
Answer relevancy	Does the answer address the question that was asked?	0.85
Context precision	Are the retrieved passages relevant, with the useful ones ranked first?	0.80
Context recall	Do the retrieved passages contain everything the reference answer needs?	0.80
Citation accuracy	Did the answer cite the page, slide or timestamp where the answer really is?	0.85
Off-material handled	Were questions outside the material declined or flagged?	0.90

5a

Standard frameworks

Choose **RAGAS**, **DeepEval** or **TruLens** per run, all judged by the local second model. Any score a framework can't produce is filled by built-in judges, and the run says how many.

5b

Team-built test set

Questions with reference answers and known source locations, plus off-material probes. Import a CSV/JSON file, or draft questions from the material and review them. A 26-question sample set is included.

5c

Simulated students

Six student profiles with hidden abilities take quizzes through the *real* quiz engine for several sessions. It reports mastery gain, adaptive vs fixed-difficulty baseline, and question repetition. Everything is rolled back afterwards.

What a run produces

- The six headline scores against their targets, with a trend across runs.
- Answer rate, average and 90th-percentile response time, and the exact models and settings used.
- Scores by source type (textbook page, slide, video timestamp, diagram) and the weakest answers with the reason they failed.
- **Per-question results:** the tutor's answer next to the reference, expected vs cited locations and each score.
- Question-bank checks (generated, verified, held back, rejected, duplicates) and the simulated-student learning curves.
- Downloadable **Markdown report** and **CSV** for sharing or submission.

Where results come from. Scores depend on the school's material and on the models running on its hardware, so they are produced by running the harness on the target machine. Development used automated tests (64 backend tests, plus browser walk-throughs of every screen) with stand-in AI responses. The first real run on the school's computer gives the actual numbers.

09 · REQUIREMENT COVERAGE

Every requirement, and where it lives

#	REQUIREMENT	HOW INTELLINOVA MEETS IT	
1a	Ingest lecture videos, textbooks and slides without manual preprocessing	Upload PDF, PPTX, video or YouTube link; six automatic stages	Done
1b	Organise into topics, concepts and prerequisites; every unit linked to page, slide or timestamp	Curriculum with prerequisites; every unit stores its origin	Done
1c	Identify topics, subtopics and key concepts; tag every unit	Tagging stage plus *Suggested topics* for new material	Done
1d	Use images, diagrams and figures	LLaVA captions and OCR labels become citable units	Done
2a	Cited excerpts that open the exact page, slide or timestamp	Per-sentence citations and a source viewer	Done
2b	Decline or flag queries the material doesn't cover	Faithfulness gate; decline, or a separately labelled outside answer	Done
3a	MCQ, short-answer and numerical quizzes and mock exams on a chosen scope, tagged by topic, source and difficulty	Practice and mock exams	Done
3b	Verify correctness; avoid repeated questions	SymPy re-solve, second-model check, de-duplication, seen-question skip	Done
3c	Cited feedback; weak-topic and misconception report	Post-quiz report with next steps	Done
4a	Per-topic mastery updated after every quiz and conversation	BKT + IRT/Elo + FSRS	Done
4b	New students with no history	Diagnostic quiz and intake conversation	Done
5a	Evaluate with RAGAS, DeepEval, TruLens or similar	All three installed; chosen per run	Done
5b	Faithfulness, relevancy, context precision and recall on a team-built test set	Test-set import or drafting; per-question results; reports	Done*
5c	Simulated students across sessions: mastery gain, repetition rate	Six profiles through the real quiz engine vs a baseline	Done
6a	Visual course flow map	Student and admin flow maps	Done
6b	Revision material for weak topics: flashcards, slides, audio	Revision packs with `.pptx` and audio brief	Done
6c	Study schedule from weak topics, forgetting curves and exam date	Study Plan	Done
6d	Mixed language: English material, Hindi explanations	Multilingual retrieval; English / Hinglish / Hindi answers	Done
6e	Audio tutoring sessions	Whisper voice input, Piper voices, hands-free mode	Done

* The metrics are produced by running the harness on the school's hardware with its test set (see section 08).

Built on open, proven components

Frontend	Next.js 15, React 19, Tailwind CSS 4, SWR. Responsive and installable on phones; light and dark themes.
Backend	FastAPI (Python 3.11), SQLAlchemy 2, Alembic migrations, Celery workers on Redis.
Data & search	PostgreSQL 16 with pgvector (HNSW) and trigram/full-text indexes; Meilisearch for the catalog.
AI (local)	Ollama (llama3.1 8B, qwen2.5 7B, llava 7B), multilingual-e5-small embeddings, mMiniLM cross-encoder, faster-whisper, Piper, Tesseract, SymPy.
Learning science	BKT (pyBKT refits), FSRS (py-fsrs), Rasch/Elo item difficulty.
Evaluation	RAGAS, DeepEval, TruLens, plus a transaction-isolated student simulator.
Delivery	Docker Compose behind Caddy (automatic HTTPS), with one-click Start / Stop / Update shortcuts for Windows.

Safety, privacy and access

- **No data leaves the machine.** Models run locally; evaluation-library telemetry is switched off.
- **Invite-only staff.** The first admin is created with a bootstrap token. Everyone else needs a single-use invite with a role: *Curator* (content) or *Admin* (also users, roles, privacy).
- **Strict role separation.** Student and staff accounts each use only their own app, enforced on the server, not just hidden in the UI.
- **Secure sessions.** GoTrue sign-in with email verification codes and httpOnly cookies. Students can request a data export, a correction or account deletion.
- **Content safety.** Auto-found videos are filtered (length, live, profanity, curator blocks) and judged against the syllabus. Unpublished content is never shown.

Running it

```
# one-time: copy .env.example to .env and fill in the secrets
docker compose --profile dev up -d --build
# first start downloads the AI models (a few GB), or double-click "Start IntelliNova" on Windows
# then open http://localhost and create the first admin at /admin/signup
```

Known limits and next steps

- **Model size.** 7–8B models fit a laptop. English answers are good; Hindi fluency and diagram reading improve with larger models (one setting) on a bigger GPU.
- **Long videos.** Lecture transcription runs on the CPU (about 15–30 minutes per hour of video); YouTube captions skip this.
- **Next:** class-level teacher dashboards per section, LMS import, and printable worksheets from the question bank.